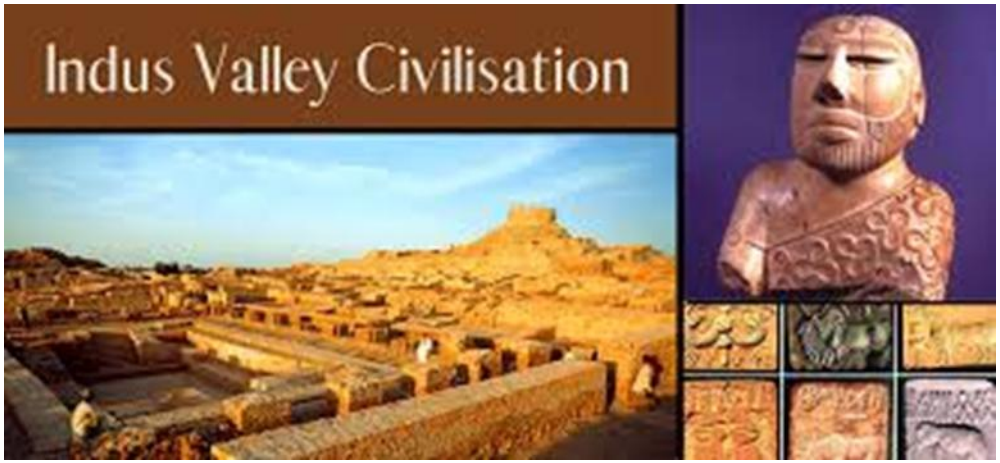
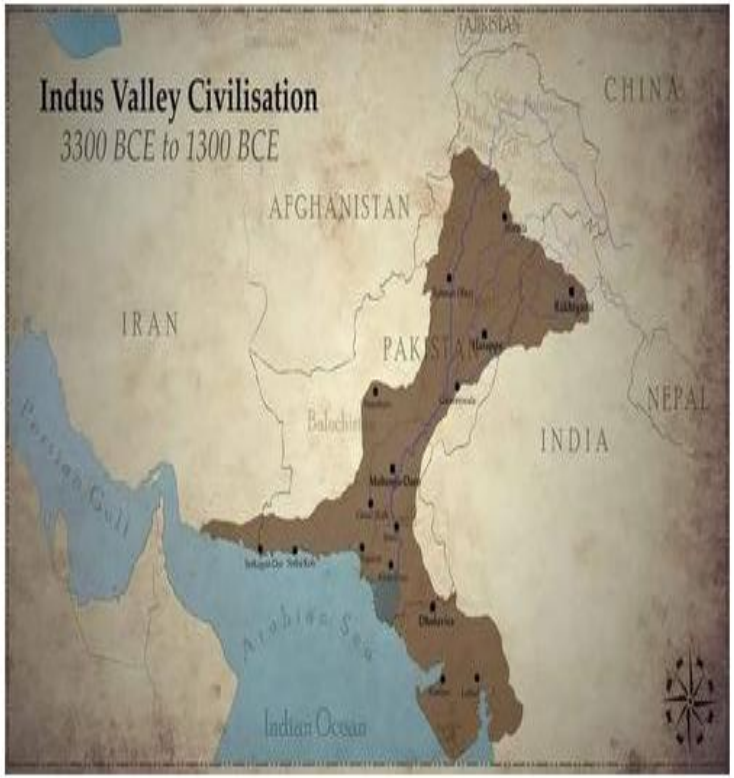
Mathematics In India
(IKS 4001)

Chapter 3: Antecedents? Mathematics in the Indus Valley

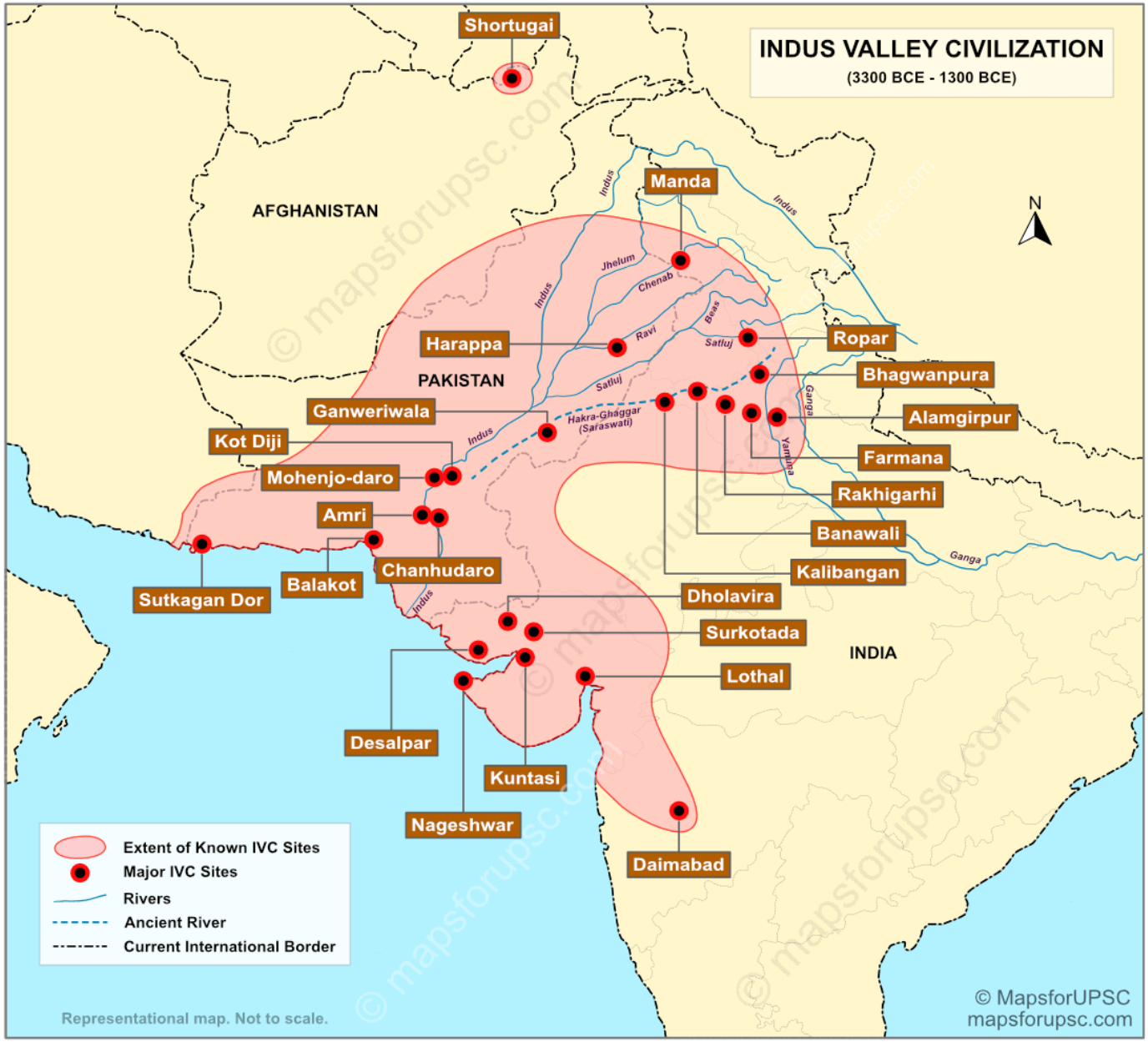


Background

- The Indus Valley Civilization (IVC) stands as one of the greatest early civilizations in human history, alongside Mesopotamia and Egypt.
- It flourished around 3300–1300 BCE in the northwestern regions of the Indian subcontinent, covering parts of present-day India and Pakistan.
- It was known for its well-planned cities, advanced drainage systems, standardized weights and measures, and thriving trade networks.
- Major sites include Harappa, Mohenjo-daro, Dholavira, and Lothal.
- The civilization mysteriously declined, but its contributions to urban planning, craftsmanship, and trade remain significant in history.



Map of IVC


INDUS VALLEY CIVILIZATION
(3300 BCE - 1300 BCE)
AFGHANISTAN
PAKISTAN
INDIA
N
Shortugai
Manda
Harappa
Ropar
Bhagwanpura
Ganweriwala
Alamgirpur
Kot Diji
Farmana
Mohenjo-daro
Rakhigarhi
Amri
Banawali
Chanhudaro
Kalibangan
Sutkagan Dor
Balakot
Dholavira
Surkotada
Lothal
Desalpar
Kuntasi
Nageshwar
Daimabad
Indus
Jhelum
Chenab
Ravi
Beas
Satluj
Hakra-Ghaggar (Saraswati)
Yamuna
Ganga
Extent of Known IVC Sites
Major IVC Sites
Rivers
Ancient River
Current International Border
Representational map. Not to scale.
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IVC – Three phases

Indus Valley Civilization (3300 to 1300 BCE)

Early Phase (3300 to 2600 BCE)

Early sites of IVC are Padri (Gujarat), Kalibangan(RJ), Dholavira(Kutch), Harappa (Pak), Balakot, Amri, and Bhirrana (Haryana), Kot Diji and Gumla

Mature Phase (2600 to 1900 BCE)

Occupied area of over 8 lakh sq km

Manda (Jammu)

Sutka-gendor (Baluchistan) — W

N

Extent

E — Alamgirpur (UP)

S

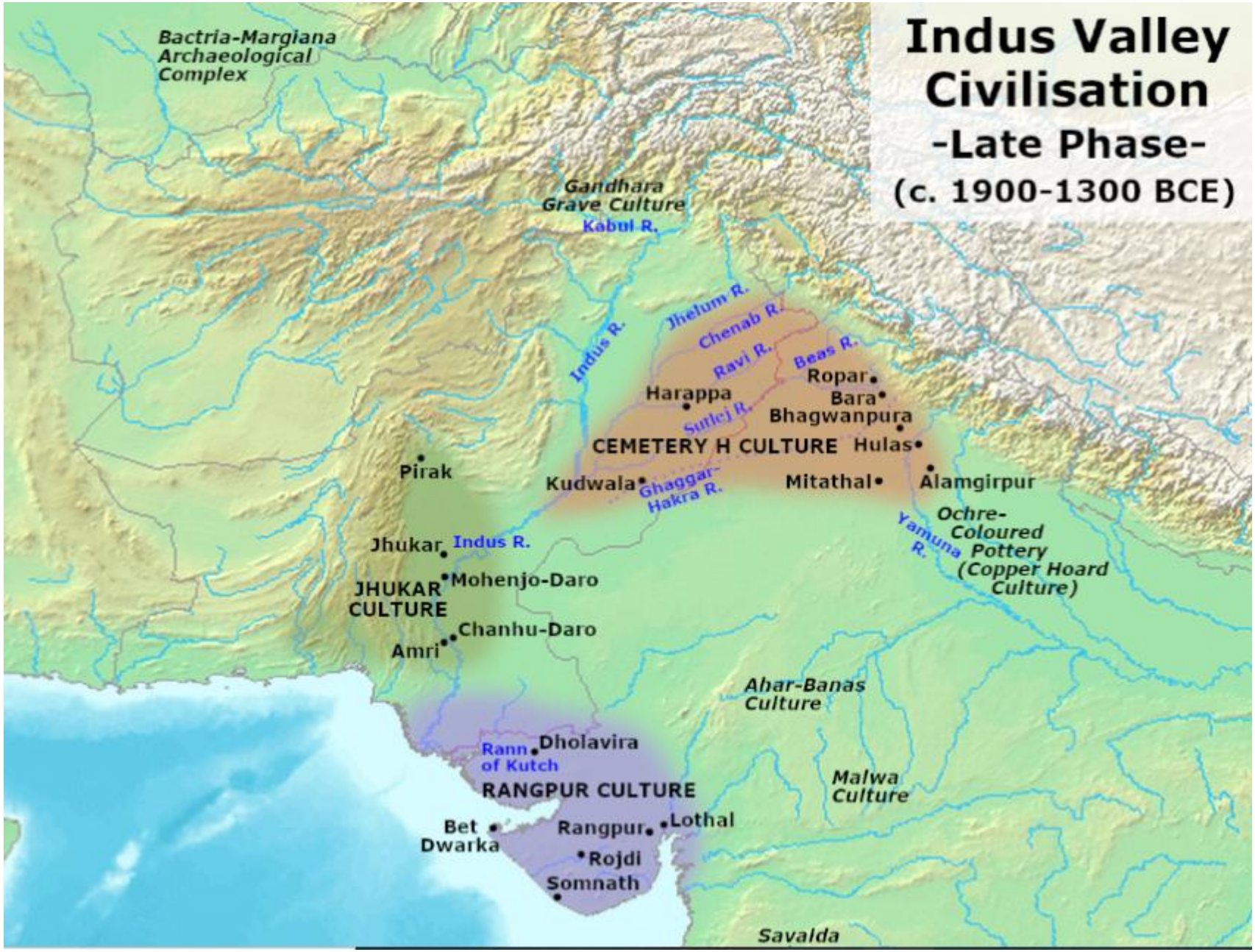
Malvan (Guj)

Late Phase (1900 to 1300 BCE)

IVC declined. Sites are more in Number but they are smaller in size. Divided into 5 phases

IVC Map – Mature phase

IVC Map – Late phase

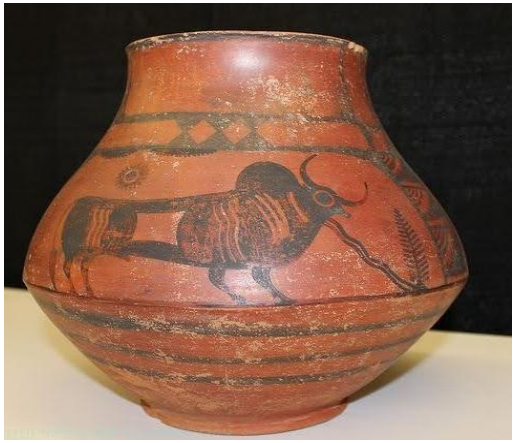


3.1 Generalities

- While the Indus Valley script remains undeciphered, the chapter argues that mathematical knowledge can be "read" through civilizational artifacts such as townscapes, street plans, and geometric decorations .
- The civilization demonstrated a high degree of standardization and technical skill, which implies an underlying mathematical culture.
- There is evidence for the use of compass like devices for drawing circles in the Indus Valley civilization

Artifacts and Cultural Treasures

Seals



Pottery

The Dancing Girl (Bronze Statue)

Jewellery & Beadwork

Indus Script & Inscriptions

Indus Valley Script

Priest King

Indus Valley Seal

Fig. 6: Seal of Indus Valley representing numerology of that period

Table 1: Symbols used in Indus Numeral System (3500 BC)

Number	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*	*13*
Indus Symbol	*I*	*II*	*III*	*IIII*	*IIII* *I*	*IIII* *II*	*IIII* *III*	*IIII* *IIII*	*IIII* *IIII* *I*	*IIII* *IIII* *II*	*IIII* *IIII* *III*	*IIII* *IIII* *IIII*	*I IIII* *IIII* *IIII*

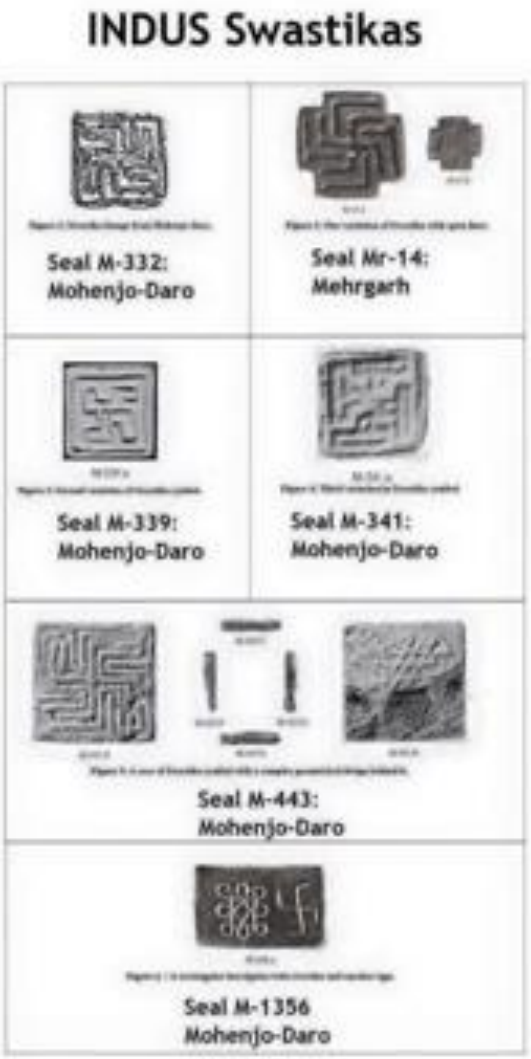
[Sandip Ghanta and Sankar Prasad Mukherjee]

Indus Seals

Pashupati Seal

Clockwise from the top : a) The Pashupati seal of Mohenjodaro; b) A seal bearing the very common unicorn motif; c) The mould of a seal showing a man fighting with two tigers ; d) An elephant in movement. All the seals have symbols from the Indus script inscribed on top.

Indus Swastikas



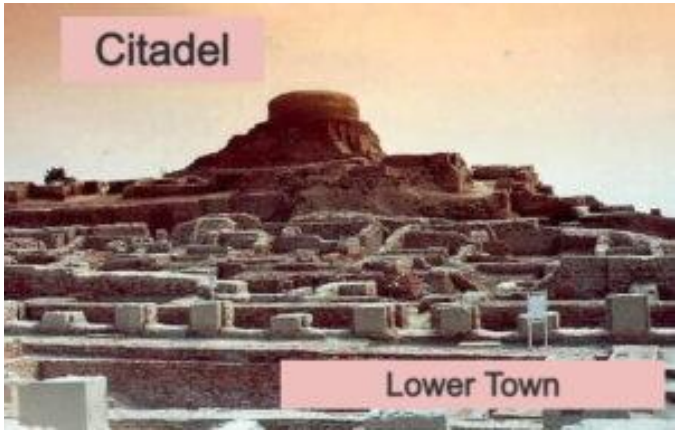
Mother Goddess , Terracota
Indus Valley

Town Planning

City Walls

Each city in the Indus Valley was surrounded by massive walls and gateways. The walls were built to control trade and also to stop the city from being flooded.

Each city had 20,000 to 40,000 people. There were 5 million individuals in Indus Valley civilization. It was 10% of the global population



The acropolis and the lower cities

- A typical city would be divided into two sections, each fortified separately
- One section was located on an artificially raised mound (sometimes called *citadel*) while the other level was on level ground
- The acropolis contained the important buildings of the city, like the assembly halls, religious structures, granaries and in the great bath in case of Mohenjo-Daro
- In-house wells almost every house had its own wells, drains and bathrooms. The in-house well is a common and recognizable feature of the Indus Valley Civilization

Town Planning

Drainage System

Each house was connected directly to an excellent drainage system, which indicates a highly developed municipal life

Granaries

The largest building found at Mohenjo-Daro is a granary, running 150 feet long, 75 feet wide and 15 feet high. The granary was divided into 27 compartments in three rows.

It was well ventilated and it was possible to fill grain in from outside. The large size of the granary probably indicates a highly developed agricultural civilization.

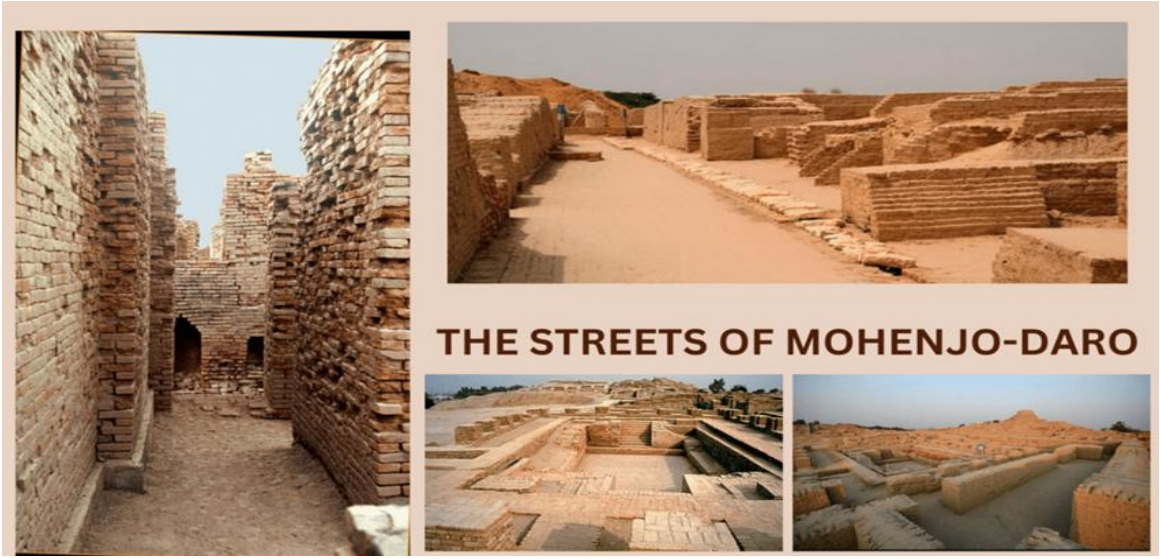
Grid Pattern

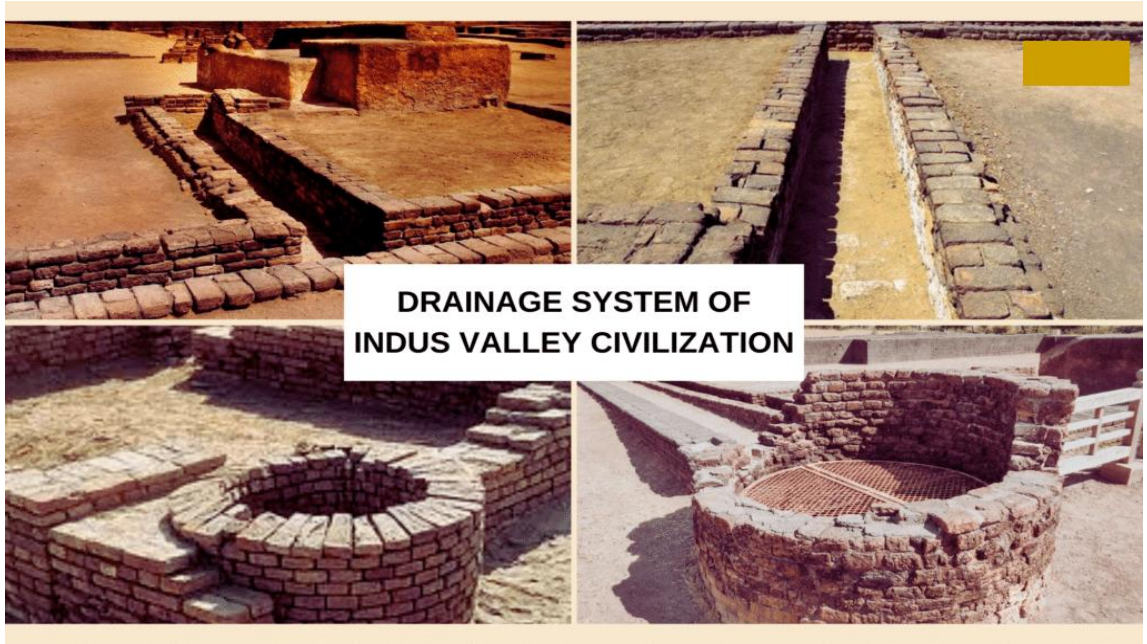
Harappa and Mohen-Jo Dero were laid out on a grid pattern and had provisions for an advanced drainage system. Streets were oriented east to west. Each street was having a well organized drainage system.

No Temples

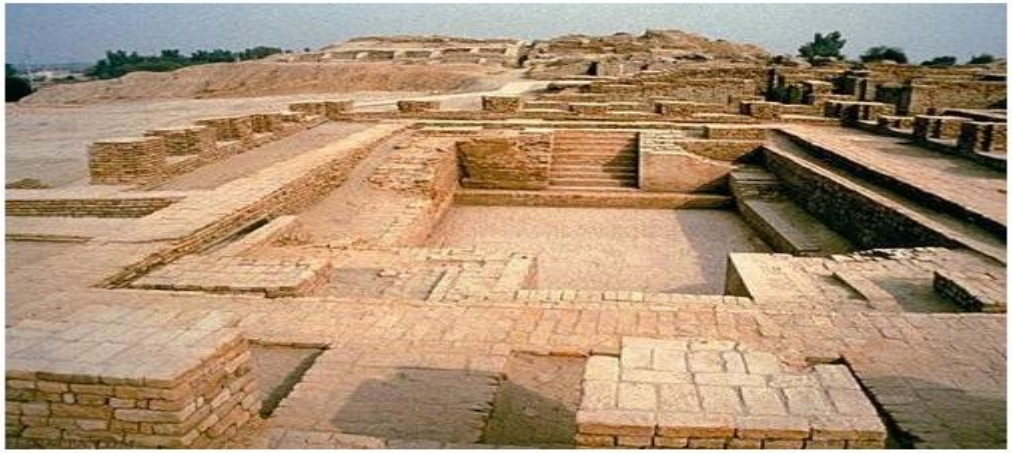
There are no traces of temple architecture or other religious places, yet the people practiced religion. The great bath has been linked to some religious practice.

Town Planning





Great Bath in Mohenjo-Daro

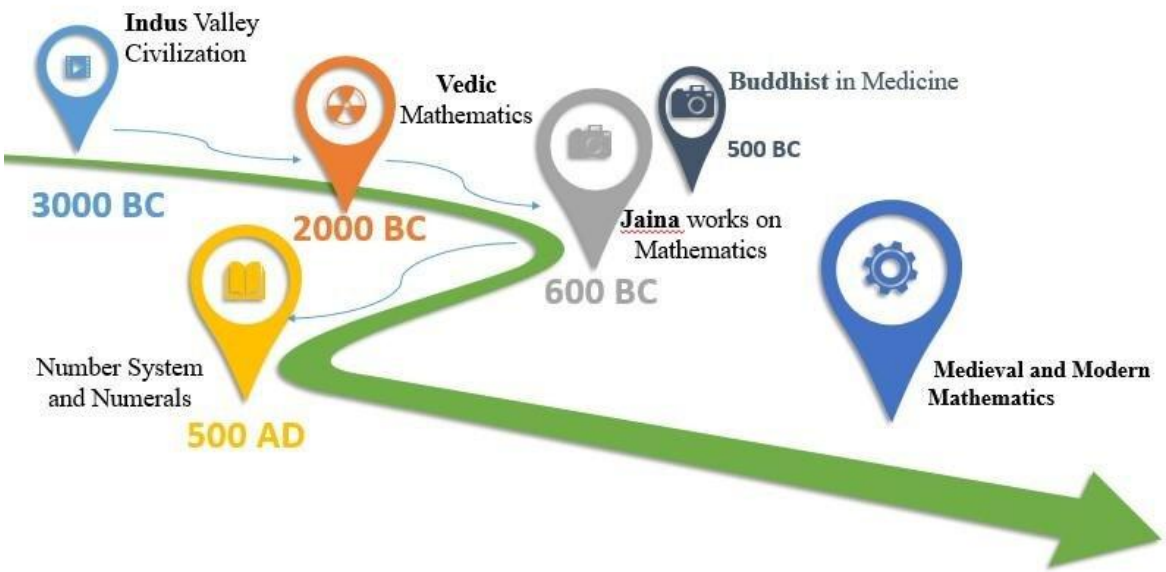


The Great bath at Mohenjo-Daro is about 179 feet long and 107 feet wide.
The complex has a large quadrangle in the center with galleries and rooms on all sides.
In the center of this quadrangle there is a large swimming enclosure that is 39 feet long, 23 feet wide and 8 feet deep.
The structure is made out of baked bricks lined with bitumen indicating that it was meant to hold water.
The entire complex is connected to an elaborate water supply and sewer system and its presence indicates some level of public management for water use and sanitation.

Mathematics in IVC

- Indus Valley civilization has no written documents but, the concept of mathematics existed at this point time.

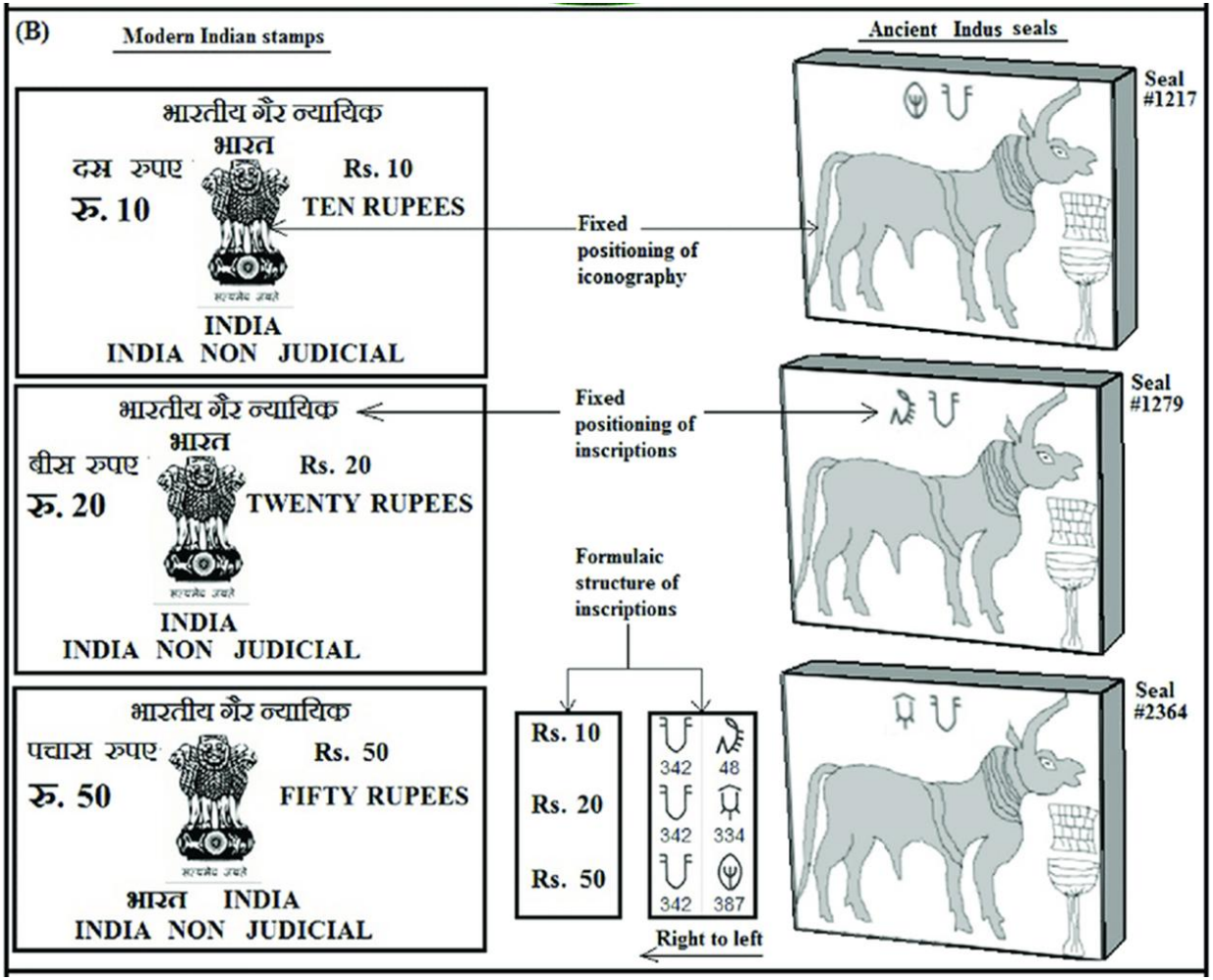
- However, following excavated remains are the proof for it e.g., civil structures, sanitary system, metalled coins and utensils, the inscription on walls, tablets and alike relics etc.



3.2 Measures and Numbers

- The people of the Indus Civilization achieved great accuracy in measuring length, mass, and time. They were among the first to develop a system of uniform weights and measures.
- Indus people would have acquired an objective sense of the passage of time from the observation of natural periodicities like the solar day and year and the phases of the moon, even though there is no surviving evidence of any such astronomical observations (unlike in Mesopotamia).
- Instruments (clocks) are specifically designed to divide the day into smaller time intervals.
- The Indus people are exceptional in the regularity of the proportions that they observed in many aspects of their activities, as was recorded by Marshall and his team.

Measures and Numbers



Harappan Bricks

- A notable feature of the Harappan culture was its extensive use of “specific geometrical shaped” kiln -fired bricks and the advanced standard of its making technology.
- In Harappa, they found 15 different sizes of rectangular piped shaped Harappan bricks with standard ratio of the three dimensions (length, breadth and thickness) of each brick was close to 4:2:1.
- Now, in the modern age this is taken into account as the optimal ratio for efficient bonding required for a powerful brick technological erection.
- It is presumed that these measurements were used in the buildings and other urban structures by the Indus people with great exactness.
- But Pythagorean theorem did not use in the dimensions of the bricks in Indus Valley.

Harappan Bricks

Wedge-shaped Bricks

- Several Indus sites also made use of wedge-shaped bricks for the building of structures with a circular plan, such as water wells.
- These are bricks whose horizontal sections are isosceles (symmetric) trapezia and are later use in the Vedic culture in the making of circular altars as well as “great altar” (mahavedi) itself had the form of a symmetric trapezium.
- Kenoyer gives the dimensions relating to one well (internal diameter of 1.2m and 36 bricks, each of length (the non parallel sides) 26 cm, covered the circumference.

Figure: A public well in Harappa

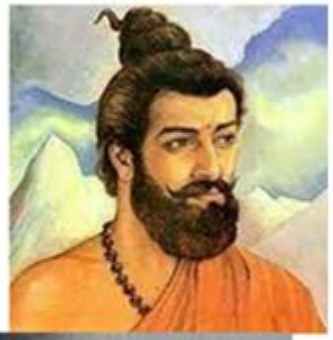
Measurement for Time, Distance, and Weight

- ✓ In the Ancient Indian literature three fundamental physical measures for quantifying length, time, and weight are found.
- ✓ There are several sources in the literature which specify units of measurements for these three physical quantities.
- ✓ For instance, in Lilavati Bhaskaracarya defined certain measures for length, volume, and mass. Arthasastra provide the units of measurement for space, time, and weight.
- ✓ Ayurvedic texts provide several measures for weight.

Notion of Paramanu

- Paramanu is a technical term in Hinduism. It is defined as the smallest and indivisible particle of matter.

- Paramanu was considered to be the smallest measure by its weight, length and time for an atomic cycle.
  - Paramanu length=2.88 x 10^(-7) mm
  - Paramanu weight= 5.79 x 10^(-5) g
  - Paramanu time= 1.31 x 10^(-5) sec



- Acharya Kanad is the founder of Vaisheshika darshan. It propounds the doctrine of paramanu (atoms) as part of its metaphysics.

- The doctrine of discovery of paramanu is somehow considered equivalent to the modern atomic theory of science.

- In modern era, Dalton is regarded to introduce paramanu to this world with his own fundamentals for defining paramanu or atom.

Measurement for Length

In Hindu scriptures, parmanu is the fundamental particle and smallest unit of length.
A **yojana** is a measure of distance that was used in ancient India.
In Arthashastra, Chanakya mentions two types of Dhanushas as units for measuring lengths and distances. One is the ordinary Dhanusha, consisting of 96 Angulas, and the other Dhanusha is mentioned as Garhpatya Dhanusha and consists of 108 Angulas, used for measurement of roads and distances. Chanakya also mentions that a Dhanurgraha consists of 4 Angulas and a Yojana consists of 8000 Dhanushas. If we keep 10 divisions of the Mohenjo-daro scale to be equal to a Dhanurgraha or 4 Angulas, the precise length of an Angula works out to be 16.764mm.

A Dhanusha of 96 Angulas = 96 x 16.764mm = 1.609344m

A **Dhanusha of 108 Angulas** = 108 x 16.764mm = 1.810512m

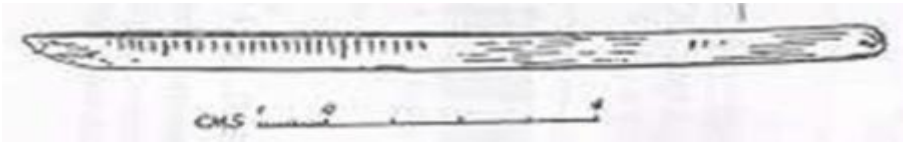
A Yojana = **8000 Dhanusha** (of 108 Angulas each) = 8000 x 1.810512m = 14.484096km

14.484096km = 9 miles, (exactly!).

1000 **Dhanushas of 96 Angulas** each = 1 mile

Indus Inch Measures and Scales

- The scales and instruments for measuring length have been excavated at major urban centres of this civilization, mainly in Mohenjo-Daro, Harappa and Lothal.
- The Mohenjo-Daro scale is a part of a scale having length of 66.2mm, with nine carefully sawn, equally spaced parallel lines, on average of 6.7056mm apart as from the excavations of the Great Bath in Mohenjo-Daro.
- One of the lines is marked by a hollow circle, and the sixth line from the circle is indicated by a large circular dot. The distance between the two markers is 1.32 inches (33.53mm), and has been named the „Indus inch".
- There are certain interesting connections/correlation between the unit of length of that time and Sumerian unit of length, called „sushi" as exactly equal to half of an „Indus inch"



Ruler found at Lothal (Indus Inch scale)

Ancient Indian Measures for Length

Unit	*Multiplier of Preceding Unit**	*No. of Paramāṇus*	*Length (in mm)*
Paramāṇu-rajas	1	1	2.8778×10^{-7}
Reṇu	7	7	2.0145×10^{-6}
Truṭi	7	49	1.4101×10^{-5}
Vātāyana-rajas	7	343	9.8709×10^{-5}
Śaśa-rajas	7	2,401	6.9096×10^{-4}
Edaka-rajas	7	16,807	4.8367×10^{-3}
Go-rajas	7	1,17,649	0.033857202
Likṣā-rajas	7	8,23,543	0.237000411
Sarṣapa	7	57,64,801	1.65900288
Yava	7	4,03,53,607	11.61302016
Aṅguli-parva	7	28,24,75,249	81.29114114

Adopted from **Ifrah, G. (2004).** *The Universal History of Numbers II*, Penguin Books, pp. 138–139.

Unit	*Multiplier of Preceding Unit**	*No. of Aṅgulas*	*Length (in metre)*
Aṅgula	1	1	0.016764
Dhanurmuṣṭi	8	8	0.134112
Prājāpatya-hasta	3	24	0.402336
Dhanus	4	96	1.609344
Gārhapatya-dhanus	1.125	108	1.810512
Goruta	2000	216,000	3621.024
Yojana	4	864,000	14484.096

* The length of an Angula is 16.764 mm as per Indus inch.

Measurement for Time

- The ancient Indians were interested in studying the origin of the Universe, which is one of the five defining characteristics of a Purana.
- The notion of time becomes very relevant in this context and measure of time become inevitable.
- Further, studies pertaining to Jyotisa (Astronomy and Astrology) also requires measures pertaining to time.
- The smallest measure of time is of the order of 10^-5 seconds and the largest number 10^11 human years.

Ancient Indian Measures for Time

Unit	Multiplier of preceding unit*	No. of Paramāṇus	Time (Seconds)
Paramāṇu	1	1	1.3133×10^{-5}
Aṇu	2	2	2.6266×10^{-5}
Trasreṇu	3	6	7.8797×10^{-5}
Truṭi	3	18	2.3639×10^{-4}
Vedha	100	1,800	2.3639×10^{-2}
Lava	3	5,400	7.0917×10^{-2}
Nimeṣa	3	16,200	0.212750617
Kṣaṇa	3	48,600	0.638251852
Kāṣṭha	5	2,43,000	3.191259259
Laghu	15	36,45,000	47.86888889
Nāḍikā	15	5,46,75,000	718.0333333
Muhūrta	2	10,93,50,000	1436.066667
Prahara	7.5	82,01,25,000	10770.51

Unit	Multiplier of preceding unit*	No. of human years
Māsa	1	0.08333
Ṛtu	2	0.16667
Ayana	3	0.50000
Human Year	2	1
Human Life Span	100	100
Celestial Life Span	360	36,000
Mahā-yuga	12000	43,20,00,000
Kalpa	1000	4,32,00,00,00,000

** For example, 100 Truṭis make a Vedha and 3 Ṛtus make an Ayana*

Measurement for Weight

- The measures of weight has a great importance in the ancient Indian time, Arthasastra gave a vivid description for various measures for weights, these measures were first formulated by the Indus valley civilization and the balances used for measuring things dates back to the Mauryan time.

- Some vivid descriptions for weights were later seen in Ayurvedic scripts and prescribed in practice are: Parmanu(smallest), Vamsi , Sarsapa , Yava, Gunja, Masaka, Karsa, Pala, Tula, Bhara(from smallest to largest at that time).

Ascending Order of Weights

- Important innovations of this civilization include standardized weights as weighting measures.
- Archaeological remains include an elaborate system of ascending weights.
- Two series were found:

I. One is present integral method 1, 2, 4, 8, 16, 32, 64 leading to decimal system of moderate time 10, 20, 40, 160, 200, 300, 640, 1600, 6400, 8000 and 12,800.

II. Equivalent weights had been in use in parts of India until recently, with conversion rates conform to the above ratios and thus help to conceptualise the basis of an elaborate system of exchange of one commodity from another.

- Such measurement system, their unit of weight is approximately 28 grams of present system which approximated to English weight “ounce”.

Harappan Weights of special geometrical measures in ascending order

Measurement for Weight

Unit	Multiplier of Preceding Unit*	No. of Paramāṇus	Weight (Grams)
Paramāṇu	1	1	5.787×10^{-5}
Vaṃśi	30	30	1.736×10^{-3}
Sarṣapa	9	270	1.563×10^{-2}
Yava	8	2,160	0.125
Guñja	4	8,640	0.5
Māṣaka	6	51,840	3
Karṣa	4	2,07,360	12
Pala	4	8,29,440	48
Tulā	100	8,29,44,000	4800
Bhāra	20	1,65,88,80,000	96000

* For example, 100 Palas make a Tulā and 4 Yavas make a Guñja

Numerical Interpretation

- Binary ratios came to be an integral part of the Indus people's handling of linear dimensions
- The mature Harappan phase that marks a widespread standardization of weights
- The binary sequence dominates the lower end of the mass scale: taking the smallest weight as the unit, the sequence doubles it at each step until we get to 64
- But, very interestingly, there is a second sequence which begins to appear at 16 and this sequence is decimal: 160 and 1600; 320 and 3200; and 640 and 6400
- This forms the Binary–decimal hybrid system

A rudimentary decimal system

- The standardized Harappa system of weight followed a dual binary decimal progression : 1, 2, 4, 8, 16, 32, 64

- Then instead of continuing with geometric progression : 160, 200, 320, 640, 1600, 3200, 6400, 8000, 12800

- Therefore tens, hundreds and thousands of previous units

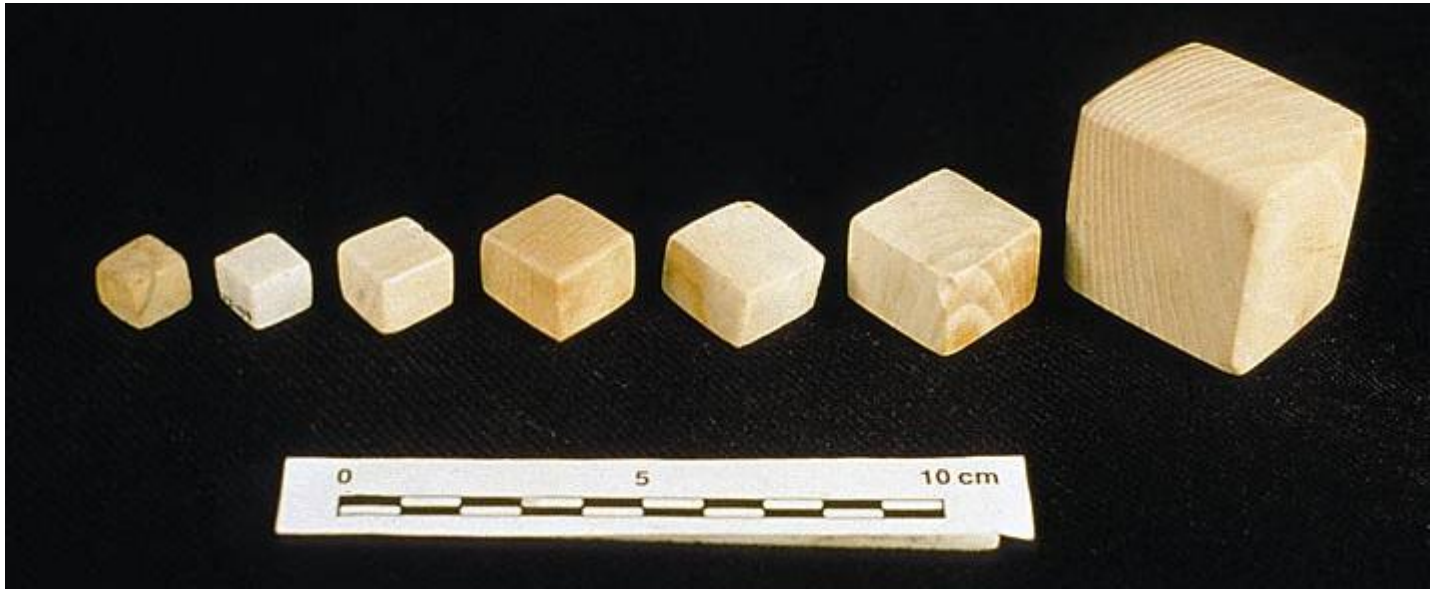


Figure : Harappan weights of special geometrical measures in ascending order

Numerical Interpretation

- Indus people counted using a base which is some power of 2, with a special role for 10
- Until decimal coinage came into force in the 1950s, Indian currency was binary: the rupee was divided into 2, 4, 8, 16 and 64 parts
- There was a proto-Dravidian language that descended from the putative Indus language and which evolved into Old Tamil and, in due course, all other Dravidian languages
- The verb root for ‘count’ in Dravidian languages is the same as the stem of the numerical noun ‘eight’
- Clearly both 8 and 10 had special significance in the counting system in Old Tamil whose vestiges are still present in modern Dravidian languages

3.3 Geometry: Overview

Two main geometric domains:

- Macro-scale geometry → Street grids, city layout (mostly right angles, straight lines)
- Micro-scale geometry → Artefacts (circles, intersecting patterns, lattices)

Transformation of Space-filling pattern in a seal excavated from Mohenjo-daro

- An example of a space-filling pattern on a seal is a Japanese fan motif utilized in various orientations to tile a square domain (Fig 1).

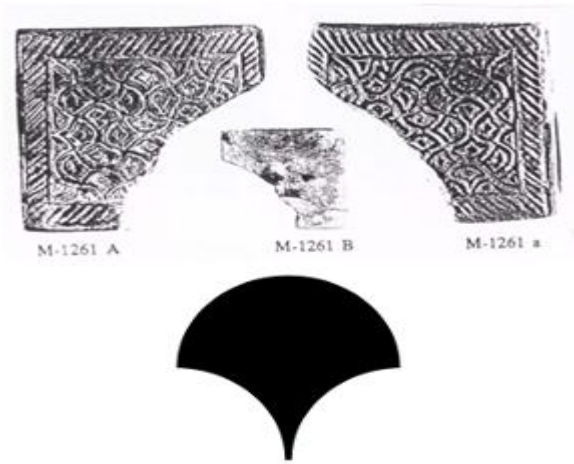
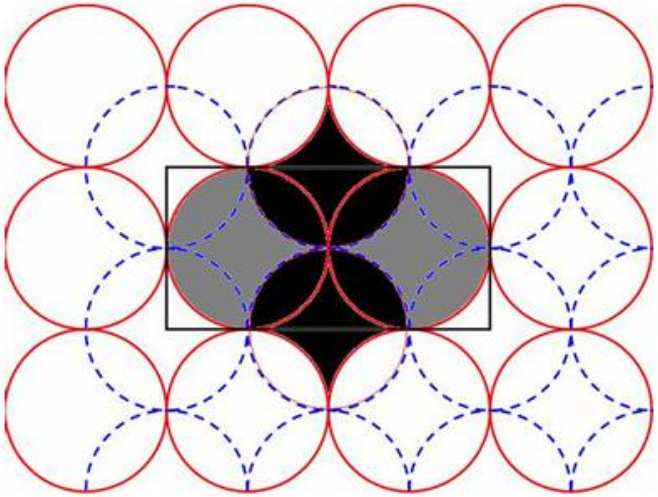


Fig 1: The seal (A), its impression (a) and the boss side (H).
On the right, we show the basic motif having the shape of a Japanese fan

Close Packing of Circles

Four of the fundamental fan-shaped motifs, each rotated by 90 degrees, make up the actual repeating pattern, which is a quite complex structure (see the shaded object in Figure 2).



- Close-packing of circles in a four-fold symmetric arrangement having two layers.
- The circles in the different layers are distinguished by broken and continuous curves, respectively.
- The two layers are displaced with respect to each other by half a lattice spacing in both the horizontal and vertical directions.
- The marked out rectangle shows the area which is transformed into the shape indicate by shaded areas.

The changes needed to create the pattern from a rectangle whose length is double its width.

- Note that, despite the complex shape of the repeating pattern, its area is fairly simple to calculate.
- By construction, this area is equal to that of the rectangle. The construction also enables us to calculate the area of the fan-shaped basic motif as being 2R^2 where R is the length from its sharply pointed tip to the midpoint of its convex edge.

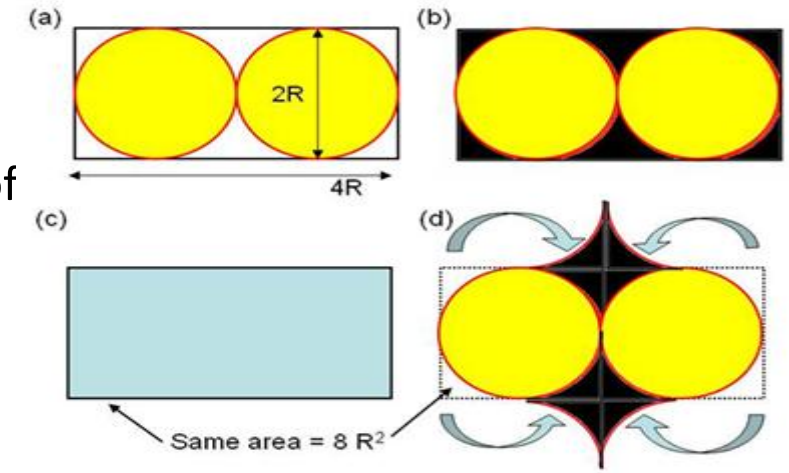
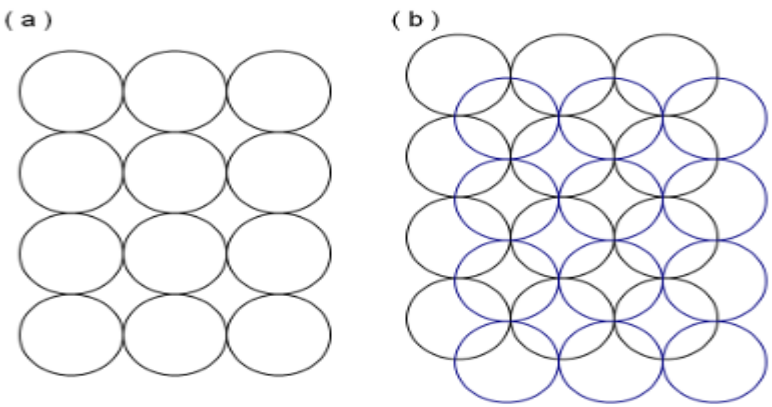


Fig 3. Two non overlapping circles, each of radius R are inscribed within a rectangle of length 4R and width 2R (a).
Of the sections of the rectangle not belonging to the circles (shaded black in (b)), four are cut and re pasted so that they fall outside the original rectangle (c), creating the repeating motif.
Note that, its area (= 8R^2) is the same as that of the original rectangle (d)

Possible variation obtained from the intersecting circle lattice

FIG. 4: (a) A periodic lattice consisting of close-packed circles in a four-fold symmetric arrangement.
(b) The intersecting circle lattice generated by superposing two lattices of circles displaced with respect to each other by half a lattice spacing in the vertical and horizontal directions.



- Fig 5 (Left) The motif of intersecting circles on incised pottery from lower levels of Mohenjo-daro.
- (Right) A decorated intersecting circle lattice used as a space filling pattern on the surface of jar from Mohenjo-daro.

Four Pointed Star and Quin-cross Sign

- If a square, the length of whose sides equal the diameter of each circle, is placed on the lattice so as to completely contain a circle inside it, one obtains a star-shaped motif by focusing exclusively within the region enclosed by the square (Fig. 6 (b)).
- A variant of this four-pointed star is obtained by relaxing the close-packed nature of the lattice so that the circles no longer touch each other.
- The resulting 'cross'-shaped motif with a small circle added at the center (Fig. 6 (d)) sometimes referred to as the Quincross sign, is found to occur very frequently in pre-Columbian iconography, especially in Mayan.

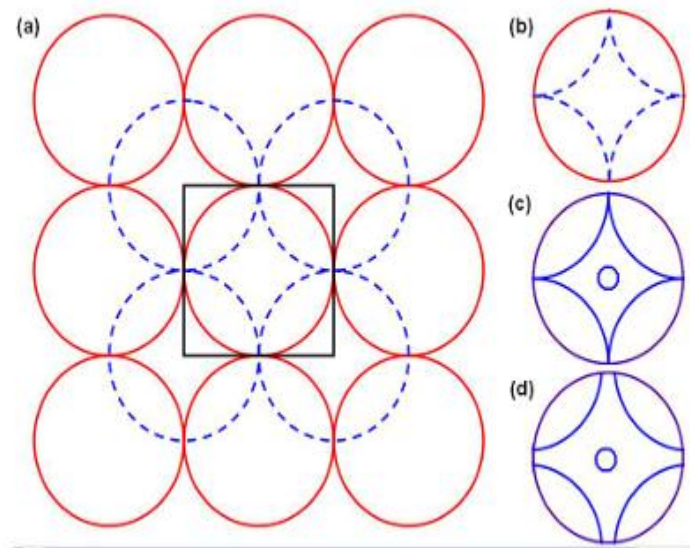
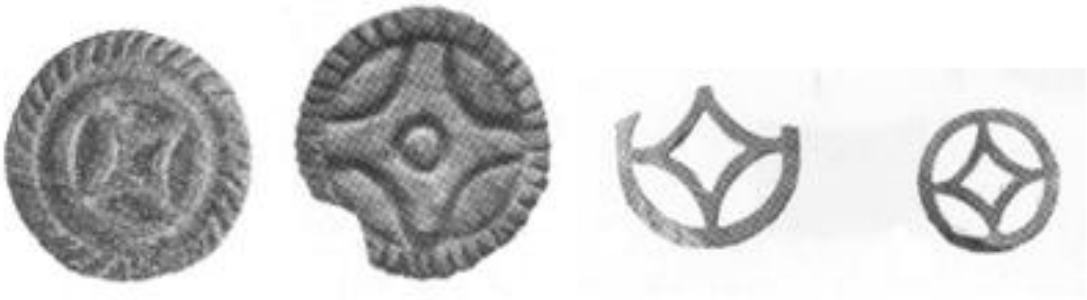


FIG. 6: The intersecting circle lattice (a) can be used to generate a star-like motif (b) with decorated variants (c) and (d).

In the Mayan context, it has been linked to Venus and may carry astral or cosmological significance.

- The motif, in both the exact four pointed star-like shape and in its cross-like variation, is found in many artifacts at both Harappa and Mohenjo-daro (Fig. 7).



- FIG. 7: The star-like geometrical motif in ornament-like objects excavated from Harappa (left) and in a seal impression (center) and ivory objects (right) found in Mohenjo-daro

3.4 Influences?

- ❖Did Indus Mathematics Influence Later India?
- ❖Chronological gap before Vedic period
  - No direct textual transmission
  - Possibility of oral/craft tradition
  - Continuity remains debated
- ❖Urban planning shows indigenous mathematical development
  - Hard to establish direct influence on Vedic geometry
  - Conceptual similarities are suggestive but not conclusive

Summary

- **Conventional view challenged**: Historians often credit the Sulbasutras as the earliest evidence of Indian geometry. The authors argue instead that Indus Valley artifacts show mathematical sophistication much earlier.
- **Evidence from artifacts**: Excavated designs reveal complex geometric motifs, especially those based on circular shapes. These include space-filling tiling patterns and intricate arrangements that suggest deliberate mathematical planning.
- **Geometric knowledge implied**: The ability to estimate areas bounded by circular arcs points to a fairly advanced understanding of geometry, not just simple decorative art.
- **Cultural continuity**: Even after the decline of Indus urban centers, this mathematical knowledge may have persisted among local populations and influenced later Vedic culture.
- **Broader significance**: The argument reframes the Indus Valley not only as technologically advanced (cities, trade, measurement systems) but also as a cradle of early mathematical thought.

Major Archaeological findings unearthed from Harappa

Probable Questions

- What sources are used to reconstruct Indus Valley mathematics?
- Explain the mathematical importance of brick ratios (1:2:4).
- What geometric features are visible in Indus town planning?
- What are the possible variation obtained from the intersecting circle lattice.
- Mention two examples of geometry from the Indus Valley.
- Did Indus Mathematics Influence Later India?